ECON 340/ENVA 300: ENVIRONMENTAL ECONOMICS (Spring 2026)

Siena University, Department of Economics

Time and venue: T/Th, 8AM-9.30AM, Siena Hall 313

Instructor:

Dr. Aakanksha Melkani; amelkani@siena.edu; (518)-783-2929
Kiernan Hall 102D
T/Th: 10.30AM–12PM; W: 11AM-1PM

Course Description: 3-credit, undergrad; This course seeks to increase understanding and critical analysis of the social and organizational causes of environmental problems. The course will suggest appropriate spheres for markets, governments, and individuals in achieving economic solutions to environmental issues.

Course requisites: None

Student learning objectives: During the semester you should develop:

1. An increased factual understanding of a range of local, regional, and international environmental issues

2. Capabilities in explaining the underlying economic reasons why environmental problems are inevitable

3. Aptitude in applying several alternative ethical criteria to suggest policy responses.

4. The application of core economic ideas to the analysis of environmental problems.

Course Materials: We will rely extensively on the on-line instructional system **Canvas** for the course material. Slides, assignments, notes, and links to readings will all be posted there. There are no required textbooks for this class, but you may find the following text useful to help you guide through more difficult concepts:

1. Environmental and Natural Resource Economics, 11th edition, Tom Tietenberg and Lynne Lewis
2. Environmental Economics – An Introduction, 6th or 7th edition, Barry C. Field and Martha K. Field
3. Principles of Microeconomics 3rd Edition by OpenStax (Free online) (https://openstax.org/details/books/principles-microeconomics-3e/?Book%20details)
4. The Economy 2.0 by the CORE (https://core-econ.org/the-economy/index.html). Free online text. You need to register at the CORE-ECON website to download this book.

Assessment and Grading

Unit	Description	Weight
Project	A self-driven research project spread over the entire semester. Research topics are chosen by the student in consultation with the instructor.	40%
In-class assignments	Small assignments that are solved in class – seeking help from classmates and instructor is allowed and encouraged.	30%
Tests	Two closed book in-class tests	20%
Participation	Subjective grade that will be assigned at the end of the semester based on your engagement and professional presence in class.	10%

Grades

Final grades will be determined based on the following break-up.

A	>= 94	B	80 to < 85	C	70 to < 72	D	58 to < 62
A-	90 to < 94	B-	75 to < 80	C-	68 to < 70	D-	50 to < 58
B+	85 to < 90	C+	72 to < 75	D+	62 to < 68	F	<50

Tentative Class Schedule

Subject to change. Any updates will be announced in class and/or via Canvas.

Week	Dates	Topics	Tasks
Part I: Using Econ concepts to understand environmental problems			
Week 1	20-Jan	Introductions, Syllabus, Class policies	Trivia
	22-Jan	Review of Econ 101: Demand, Supply, Prices, Markets, Welfare	Assign reading
Week 2	27-Jan	Review of Econ 101: Price elasticity; Taxes and Subsidies,	Class discussions based on reading
	29-Jan	Pareto optimality; Welfare theorems; Efficiency vs Equity	
Week 3	3-Feb	Failure of welfare theorems - externalities and property rights	Discussion on class project; share meeting time slots
	5-Feb	Symposium	Attend symposium and submit written assignment
Week 4	10-Feb	Externalities: Dams	Share groups and project topics in class; Select a slot for meeting
	12-Feb	Incomplete property rights: Water rights	
Week 5	17-Feb		Review; Discussion of project if needed
	19-Feb		In-class test
Week 6	24-Feb		In-person meetings
	26-Feb		In-person meetings
Week 7	3-Mar	Spring Break- no classes	Work on project synopsis (I will share outline)
	5-Mar		
PART II: Economic policy to combat environmental issues			
Week 8	10-Mar 12-Mar	Introduction to environmental policies Taxes/Tolls: Gasoline taxes, Congestion pricing Subsidies: Clean energy	Submit project synopsis
Week 9	17-Mar 19-Mar	Taxes and subsidies continued	Presentation of synopsis in class; Peer evaluation
Week 10	24-Mar 26-Mar	Tradeable permits: California Cap and Trade program, will it work in New York?	Work on project draft
Week 11	31-Mar 2-Apr	Command and Control - The Clean Air Act No class on 2nd-April	Assign peer reviews. Submit draft for peer review.
Week 12	7-Apr 9-Apr	Decentralized policy: Common Law - Flint water crisis; Statutory Law: Superfund	Peer reviews due
Week 13	14-Apr 16-Apr	Comparison of different policies	Work on final project presentation
Week 14	21-Apr 23-Apr		Review In-class test

Week 15	28-Apr 30-Apr	Final Project presentations

Communication

Apart from the classroom instruction, I will extensively use your Siena email id and Canvas to communicate with you. **PLEASE check your email and Canvas regularly for new communication and announcements.** You are encouraged to communicate and seek guidance with the instructor, especially during the office hours. You can also communicate through email and setup appointments if the office hours do not work for you. I will try to reply to all emails within 24-hours during working hours (Monday to Friday, 8AM to 5PM). My responses during other times may be slow so please plan your work ahead and do not keep urgent questions for the weekend.

Class Policy around the use of Artificial Intelligence tools

You may use all online resources accessible to you to help you understand the class material. **However, all assignment submissions should be your own individual effort.** Copying and pasting text from online resources, including Chat-GPT, can be considered plagiarism, which is a case of academic dishonesty. Please read https://www.siena.edu/offices/academics/academic-policies-and-procedures/academic-integrity-policy/ to familiarize yourself with the College academic integrity regulations

Attendance and leave policy

If you miss a class, it is your responsibility to cover the missed class material. You may re-take any missed in-class assignments, but you will NOT receive a grade for those. Generally, no leaves are allowed on the day of an exam unless there is an exceptional unforeseen reason for the absence and/or if you have taken prior permission from the instructor. If you miss an exam with an unexcused absence, you will receive a zero for the exam. If you miss an exam for an unforeseen reason, you should contact me as soon as possible. In case of long-term absence (more than a week) please refer to Siena's leave policy. For more details on university policies please see: https://www.siena.edu/offices/health-services/medical-leave/

Make-up exams

Make-up exams are given only for excused absences and with the instructor's prior approval. If you miss an exam with an unexcused absence, you will receive a zero for the exam. If you miss an exam for an unforeseen reason, you should contact me as soon as possible. If there has been an extraordinary reason for excessive absences, such as prolonged illness, a student may petition the Vice President for Academic Affairs in writing to consider reinstatement in the class or permission to withdraw with the grade W. It is the personal responsibility of the student to make up all work assigned during an absence from any class or lab.

Late assignments

Work which is late will receive grade reductions unless prior permission to submit after the due date is obtained.